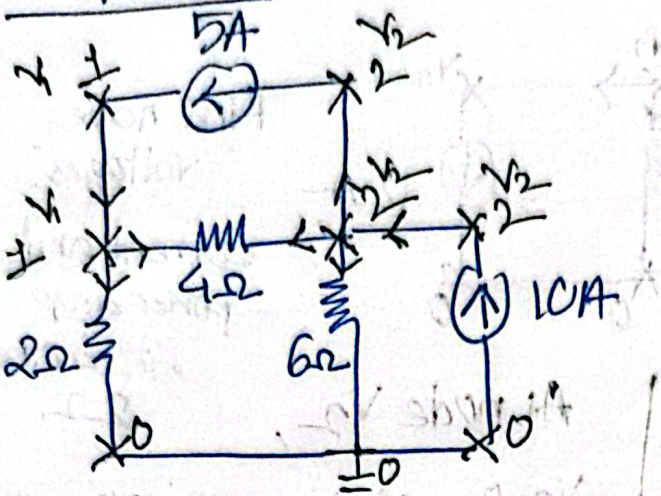
Nodal Analysis

example 3.1



Calculate Node Voltages.
Find current through 4Ω resistor. Power dropat 6Ω.

In node 1,

$$\frac{V_1 - V_2}{4} + \frac{V_1 - 0}{2} = 5$$

$$\Rightarrow 2V_1 - 2V_2 + V_1 - V_2 + 2V_1 = 20$$

$$\Rightarrow \boxed{5V_1 - V_2 = 20} \quad \text{(i)}$$

In node 2,

$$\frac{V_2 - V_1}{4} + \frac{V_2 - 0}{6} + 10 = 5$$

$$\Rightarrow 6V_2 - 6V_1 + 4V_2 + 120 = 240$$

$$\Rightarrow -6V_1 + 10V_2 = 120$$

$$\Rightarrow \boxed{-3V_1 + 5V_2 = 60} \quad \text{(ii)}$$

solving eqn (i) and (ii),

$V_1 = 13.33V$

$V_2 = 20V.$

current through 4Ω resistor,

$$I_{4\Omega} = \frac{V_1 - V_2}{4} = \frac{13.33 - 20}{4} = -1.667A$$

→ $V_2 - V_1$ or $V_1 - V_2$ এভাবে বসালেও হবে।

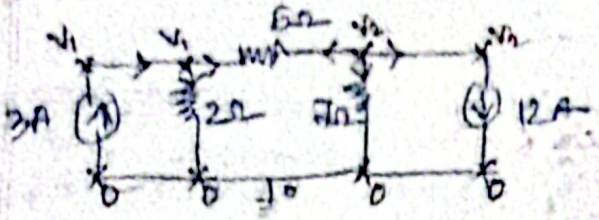
$P = \frac{V^2}{R} = I^2R$

power dropat through 6Ω resistor, $\left[I_{6\Omega} = \frac{V_2 - 0}{6} = \frac{20}{6}A\right]$

$$P_{6\Omega} = (I_{6\Omega})^2 \times 6 = \left(\frac{20}{6}\right)^2 \times 6 = 66.67W.$$

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Practice Problem 3.1



Find node voltages current and power drop at 6 and 2Ω.

At node V_1,

$$\frac{V_1-V_2}{6}+\frac{V_1-0}{2}=3$$

$$\Rightarrow V_1-V_2+3V_1=18$$

$$\Rightarrow \boxed{4V_1-V_2=18} \quad \text{—(i)}$$

At node V_2,

$$\frac{V_2-0}{7}+\frac{V_2-V_1}{6}+12=0$$

$$\Rightarrow 6V_2+7V_2-7V_1+504=0$$

$$\Rightarrow -7V_1+13V_2=-504$$

$$\Rightarrow \boxed{7V_1-13V_2=504} \quad \text{—(ii)}$$

Solving eqn. (i) and (ii),

$V_1 = -6V$

$V_2 = -42V$

current through 6Ω,

$$I_{6\Omega}=\frac{V_1-V_2}{6}=\frac{-6-(-42)}{6}=\frac{-6+42}{6}=\frac{36}{6}=6A$$

power drop,

$$P_{6\Omega}=(I_{6\Omega})^2\times 6=6^2\times 6=216W$$

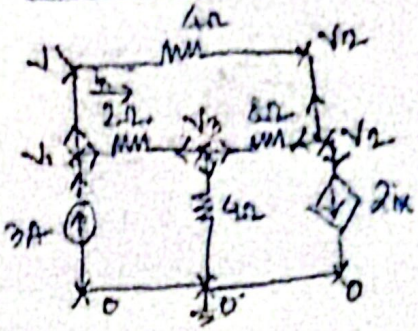
current through 2Ω,

$$I_{2\Omega}=\frac{V_1-0}{2}=\frac{-6-0}{2}=-3A$$

power drop,

$$P_{2\Omega}=(I_{2\Omega})^2\times 2=(-3)^2\times 2=9\times 2=18W$$

Example 3.2



Find node voltages, power drop at 2Ω, voltage drop at 6Ω, current through it.

In node V_1,

$$\frac{V_1-V_3}{2}+\frac{V_1-V_2}{4}=3$$

$$\Rightarrow 2V_1-2V_3+V_1-V_2=12$$

$$\Rightarrow \boxed{3V_1-V_2-2V_3=12} \quad \text{—(i)}$$

In node V_2,

$$\frac{V_2-V_1}{4}+\frac{V_2-V_3}{6}+2ix=0$$

$$\Rightarrow 2V_2-2V_1+V_2-V_3-16ix=0$$

$$\Rightarrow \boxed{-2V_1+3V_2-V_3-16ix=0} \quad \text{—(ii)}$$

in node V_3,

$$\frac{V_3-V_1}{2}+\frac{V_3-0}{4}+\frac{V_3-V_2}{6}=0$$

$$\Rightarrow 6V_3-6V_1+12V_3+2V_3-2V_2=0$$

$$\Rightarrow \boxed{-6V_1-2V_2+20V_3=0} \quad \text{—(iii)}$$

$$i_x=\frac{V_1-V_3}{2}$$

$$16ix=8(V_1-V_3)=8V_1-8V_3$$

putting value in eqn (ii),

$$-2V_1+3V_2-V_3+8V_1-8V_3=0$$

$$\Rightarrow \boxed{6V_1+3V_2-9V_3=0} \quad \text{—(iv)}$$

Solving eqn (i), (ii), (iii) we get,

$V_1 = 4.8V$

$V_2 = -2.4V$

$V_3 = 2.4V$

power drop at 2Ω,

$P_{2\Omega} = (I_{2\Omega})^2 \times 2$

$= (1.2)^2 \times 2$

$= 2.88W.$

$I_{2\Omega} = \frac{V_1 - V_3}{2}$

$= \frac{4.8 - 2.4}{2}$

$= 1.2A$

voltage drop at 6Ω,

$V_{6\Omega} = V_3 - V_2$

$= 2.4 - (-2.4)$

$= 2.4 + 2.4$

$= 4.8V.$

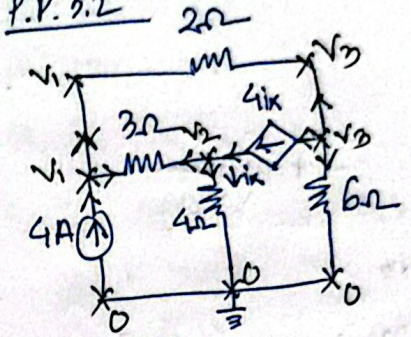
$(V_2 - V_3)$ can also be done

current i_x,

$i_x = \frac{V_1 - V_2}{2\Omega} = I_{2\Omega}$

$= 1.2A.$ LSA

P.P. 3.2



Find node voltages.

In node V_1,

$\frac{V_1 - V_2}{3} + \frac{V_1 - V_3}{2} = 4$

$\Rightarrow 2V_1 - 2V_2 + 3V_1 - 3V_3 = 24$

$\Rightarrow \boxed{5V_1 - 2V_2 - 3V_3 = 24}$ ——(i)

In node V_3,

$\frac{V_3 - 0}{6} + 4i_x + \frac{V_3 - V_1}{2} = 0$

$\Rightarrow \frac{V_3}{6} + V_2 + \frac{V_3 - V_1}{2} = 0$

$\Rightarrow V_3 + 6V_2 + 3V_3 - 3V_1 = 0$

$\Rightarrow \boxed{-3V_1 + 6V_2 + 4V_3 = 0}$ ——(iii)

In node for i_x,

$i_x = \frac{V_2 - 0}{4} = \frac{V_2}{4}$

$4i_x = \frac{4V_2}{4} = V_2$

In node V_2,

$\frac{V_2 - V_1}{3} + \frac{V_2}{4} = 4i_x$

$\Rightarrow 4V_2 - 4V_1 + 3V_2 = 12 \times 4i_x$

$\Rightarrow -4V_1 + 7V_2 = 12 \times V_2$

$\Rightarrow \boxed{-4V_1 - 5V_2 = 0}$ ——(ii)

solving eqn (i), (ii) and (iii) we get,

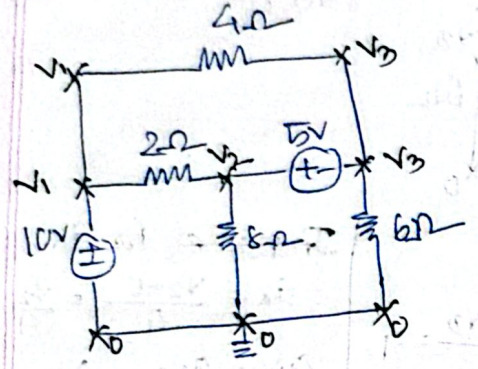
$V_1 = 32V.$

$V_2 = -25.6V$

$V_3 = 62.4V.$

Supernode math

example



Find node voltages.

In node 1, a voltage source is connected directed.

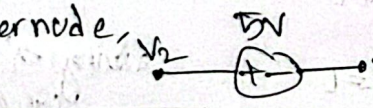
$\therefore V_1 - 0 = +10V$ (V1 এর দিকে +)

$\Rightarrow \boxed{V_1 = 10V.}$ — (i)

(node value পেলাম 2 node এ equation পাওয়া যাবে).

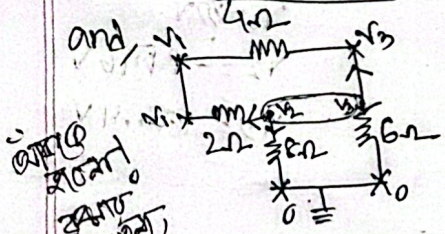
V_2, V_3 এর মধ্যে supernode আছে।

for supernode,



$\boxed{V_2 - V_3 = 5}$ — (ii)

and,



এখানে KCL প্রয়োগ করে পাওয়া যাবে।

Again,

$\frac{V_2 - 0}{8} + \frac{V_2 - V_1}{2} + \frac{V_3 - 0}{6} + \frac{V_3 - V_1}{4} = 0$

$\Rightarrow 3V_2 + 12V_2 - 12V_1 + 4V_3 + 6V_3 - 6V_1 = 0$

$\Rightarrow -18V_1 + 15V_2 + 10V_3 = 0$

$\Rightarrow -18(10) + 15V_2 + 10V_3 = 0$

$\Rightarrow 15V_2 + 10V_3 = 180$

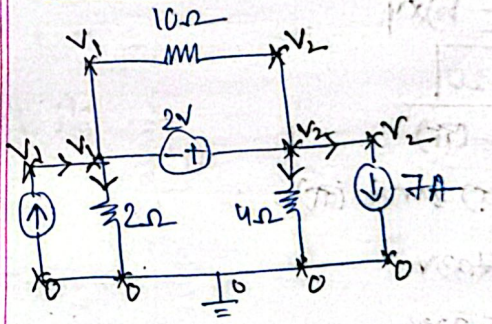
$\Rightarrow \boxed{3V_2 + 2V_3 = 36}$ — (iii)

Solving eqn (ii) and (iii),

$V_2 = 9.2V$

$V_3 = 4.2V$

Example 3.3

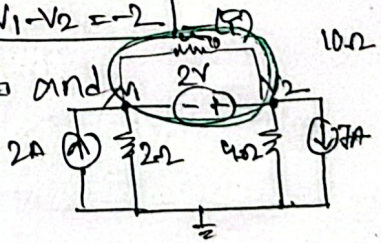


Find node voltages.

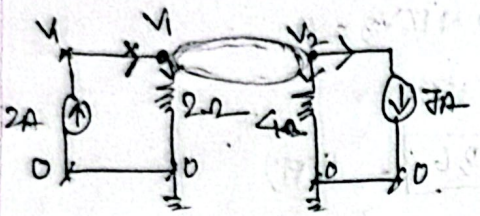
In case of supernode,

$\boxed{V_1 - V_2 = -2}$ (i)

and



10Ω এর current কোথায় চলে গেল? এটি supernode এর connection. supernode এ started and ended, so we don't count it while making an equation.



So, $\frac{V_1-0}{2}+\frac{V_2-0}{4}$ ~~or~~ $+7=2$

$\Rightarrow \frac{2V_1+V_2}{4}=2-7$

$\Rightarrow 2V_1+V_2=(-5)\times 4$

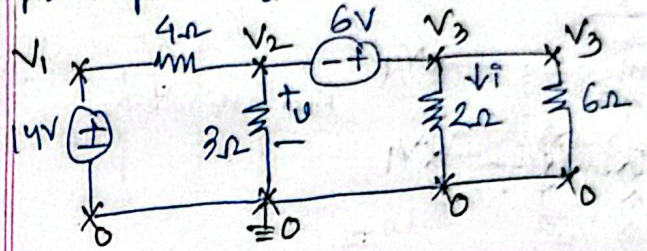
$\therefore \boxed{2V_1+V_2=-20}$ ——(i)

Solving equation (i) and (ii),

$v_1 = -7.33V$

$v_2 = -5.33V$.

practice problem 3.3



Find v and i.

For node 1,

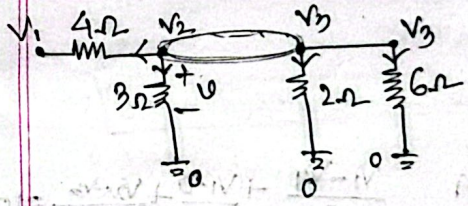
$V_1-0=14$

$\Rightarrow V_1=14$

For ~~node~~ supernode,

$\boxed{V_2-V_3=-6}$ ——(i)

For supernode,



$\frac{V_2-0}{3}+\frac{V_2-V_1}{4}+\frac{V_3-0}{2}+\frac{V_3-0}{6}=0$

$\Rightarrow \frac{V_2}{3}+\frac{V_2-14}{4}+\frac{V_3}{2}+\frac{V_3}{6}=0$

$\Rightarrow \frac{4V_2+3V_2-42+6V_3+2V_3}{12}=0$

$\Rightarrow \boxed{7V_2+8V_3=42}$ ——(ii)

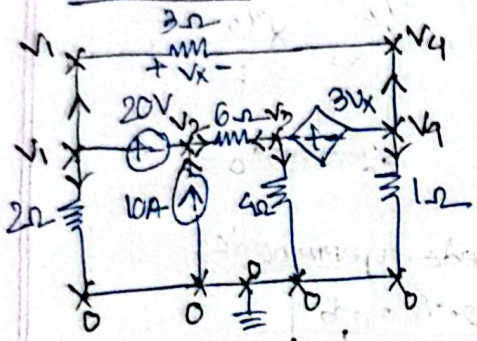
Solving eq(i) and (ii),

$V_2 = -0.4V, V_3 = 5.6V$

$v = V_2-0 = -0.4-0 = -0.4V$.

$i = \frac{V_3-0}{2} = \frac{5.6-0}{2} = 2.8A$.

Example 3.4



Find node voltages.

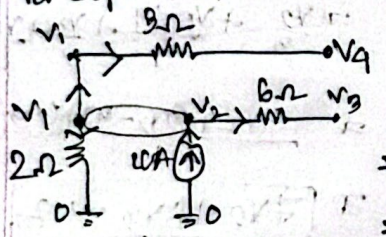
For V_x,

$V_1 - V_4 = V_x$

at supernode between V_1 and V_2,

$$\boxed{V_1 - V_2 = 20} \quad \text{—(i)}$$

For super node,



$$\frac{V_1 - V_4}{3} + \frac{V_1 - 0}{2} + \frac{V_2 - V_3}{6} = 10$$

$$\Rightarrow 2V_1 - 2V_4 + 3V_1 + V_2 - V_3 = 60$$

$$\Rightarrow \boxed{5V_1 + V_2 - V_3 - 2V_4 = 60} \quad \text{—(ii)}$$

At For super node between V_3 and V_4,

$V_3 - V_4 = 3V_x$

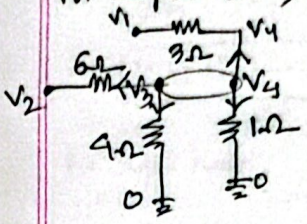
$\Rightarrow V_3 - V_4 = 3(V_1 - V_4)$ — দিয়ে হলে value বসিয়ে দিবে

$\Rightarrow V_3 - V_4 = 3V_1 - 3V_4$

$\Rightarrow 3V_1 - 3V_4 - V_3 + V_4 = 0$

$\therefore \boxed{3V_1 - V_3 - 2V_4 = 0}$ —(iii)

For super node,



$$\frac{V_3 - 0}{4} + \frac{V_3 - V_2}{6} + \frac{V_4 - 0}{1} + \frac{V_4 - V_1}{3} = 0$$

$$\Rightarrow 3V_3 + 2V_3 - 2V_2 + 12V_4 + 4V_4 - 4V_1 = 0$$

$$\Rightarrow \boxed{-4V_1 - 2V_2 + 5V_3 + 16V_4 = 0} \quad \text{—(iv)}$$

now,

$V_1 = 20 + V_2$

so, eqn (ii) becomes,

$5(20 + V_2) + V_2 - V_3 - 2V_4 = 60$

$\Rightarrow 100 + 5V_2 + V_2 - V_3 - 2V_4 = 60$

$\Rightarrow \boxed{6V_2 - V_3 - 2V_4 = -40} \quad \text{—(v)}$

eqn (iii) becomes,

$3(20 + V_2) - V_3 - 2V_4 = 0$

$\Rightarrow 60 + 3V_2 - V_3 - 2V_4 = 0$

$\Rightarrow \boxed{3V_2 - V_3 - 2V_4 = -60} \quad \text{—(vi)}$

eqn (iv) becomes,

$-4(20 + V_2) - 2V_2 + 5V_3 + 16V_4 = 0$

$\Rightarrow -80 - 4V_2 - 2V_2 + 5V_3 + 16V_4 = 0$

$\Rightarrow \boxed{-6V_2 + 5V_3 + 16V_4 = 80} \quad \text{—(vii)}$

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Solving eqn (v), (vi), (vii),

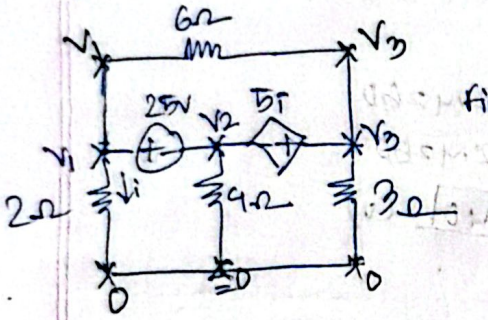
$V_2 = 6.67V$

$V_3 = 17.33V$

$V_4 = -46.67V$.

$V_1 = 20 + V_2 = 20 + 6.67 = 26.67V$.

practice problem 3.4



find node voltages V_1, V_2, V_3 and i.

supernode between V_1 and V_2,

$$V_1 - V_2 = 25 \quad \text{(i)}$$

for supernode between V_2 and V_3,

$V_2 - V_3 = -5i$

Again, $i = \frac{V_1 - 0}{2} = \frac{V_1}{2}$

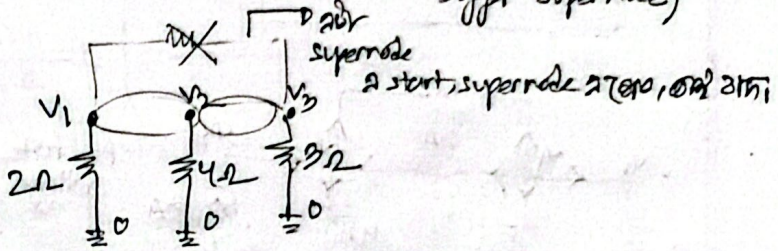
$5i = \frac{5V_1}{2}$

so, $V_2 - V_3 = -\frac{5V_1}{2}$

$\Rightarrow 2V_2 - 2V_3 = -5V_1$

$\Rightarrow \boxed{5V_1 + 2V_2 - 2V_3 = 0}$ (ii)

for supernode, (here 2 supernodes merge and make a bigger supernode)



$$\therefore \frac{V_1 - 0}{2} + \frac{V_2 - 0}{4} + \frac{V_3 - 0}{3} = 0$$

$\Rightarrow \boxed{6V_1 + 3V_2 + 4V_3 = 0}$ (iii)

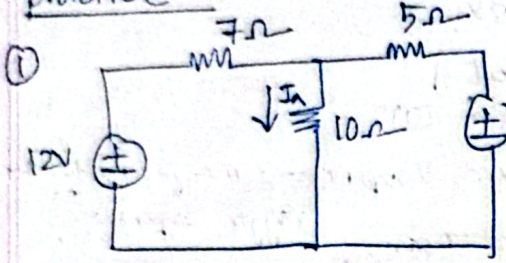
solving eqn (i), (ii), (iii) we get,

$V_1 = 7.61V$

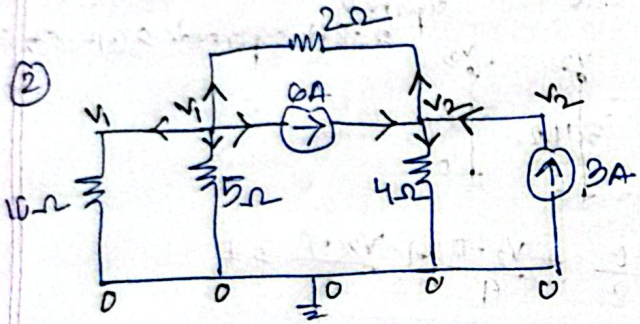
$V_2 = -17.39V$

$V_3 = 1.63V$

practice

① 

Find I_x

② 

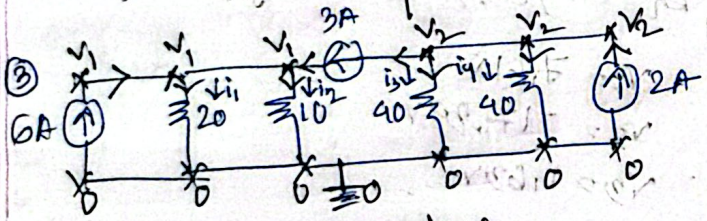
Find node voltages.

Hint: For node V_1,

$$\frac{V_1}{10}+\frac{V_1}{5}+\frac{V_1-V_2}{2}+6=0$$

For node V_2,

$$\frac{V_2}{4}+\frac{V_2-V_1}{2}=6+3$$

③ 

Find i_1, i_2, i_3, i_4.

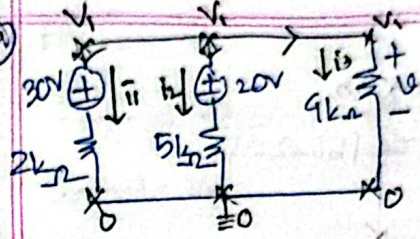
Hint: For node V_1,

$$\frac{V_1}{20}+\frac{V_1}{10}=6+3$$

For node V_2,

$$\frac{V_2}{40}+\frac{V_2}{40}+2+3=2$$

$$i_1=\frac{V_1-0}{20},\ i_2=\frac{V_1-0}{10},\ i_3=\frac{V_2-0}{40},\ i_4=\frac{V_2-0}{40}$$

④ 

Find v_o, use nodal analysis.

Sol^n:

at node V_1,

$$\frac{V_1-30}{2k\Omega}=i_1 \quad \Big| \quad \frac{V_1-20}{5k}=i_2 \quad \Big| \quad i_3=\frac{V_1-0}{4k}=\frac{V_1}{4k}$$

$$i_1+i_2+i_3=0$$

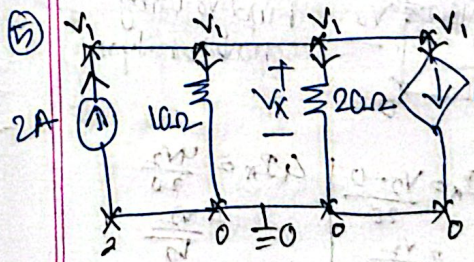
$$\Rightarrow \frac{V_1-30}{2000}+\frac{V_1-20}{5000}+\frac{V_1}{4000}=0$$

$$\Rightarrow \frac{10V_1-300+4V_1-80+5V_1}{20000}=0$$

$$\Rightarrow 19V_1=380$$

$$\therefore V_1=\frac{380}{19}=20V.$$

$$V_o=V_1-0=20-0=20V.$$

⑤ 

Find V_x.

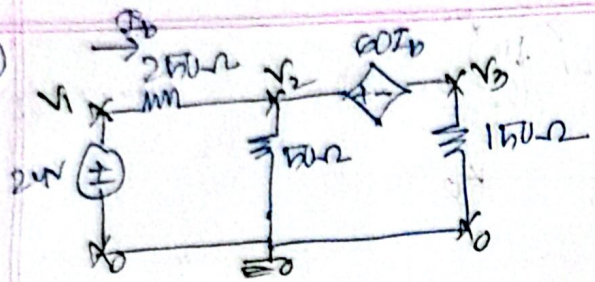
hint: $V_x=V_1-0=V_1$

and,

$$\frac{V_1}{10}+\frac{V_1}{20}+0.2V_x=2$$

$$\Rightarrow \frac{V_1}{20}+\frac{V_1}{20}+0.2V_1=2$$

⑥



Hint

$V_1 = 24V$.

supernode, $V_2 - V_3 = 60I_b$

$I_b = \frac{V_1 - V_2}{250}$

$\Rightarrow V_2 - V_3 = \frac{60(V_1 - V_2)}{250}$

$\Rightarrow 25V_2 - 25V_3 = 6V_1 - 6V_2$

$\Rightarrow 6V_1 - 31V_2 + 25V_3 = 0$

$\Rightarrow 6 \times 24 - 31V_2 + 25V_3 = 0$

$\Rightarrow 31V_2 - 25V_3 = 144$

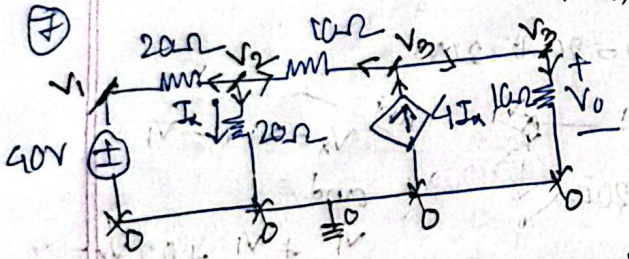
And supernode,



$\frac{V_2}{150} + \frac{V_2 - V_1}{250} + \frac{V_3}{150} = 0$

$\Rightarrow \frac{V_2}{150} + \frac{V_2 - 24}{250} + \frac{V_3}{150} = 0$

⑦



Find V_o using node analysis.

$V_o = V_3 - 0$

$\therefore V_o = V_3$

$V_1 - 0 = 40$

$\therefore V_1 = 40$

$I_x = \frac{V_2 - 0}{20} = \frac{V_2}{20}$

$4I_x = \frac{4V_2}{20} = \frac{V_2}{5}$

for node V_2,

$\frac{V_2 - V_1}{20} + \frac{V_2 - 0}{20} + \frac{V_2 - V_3}{10} = 0$

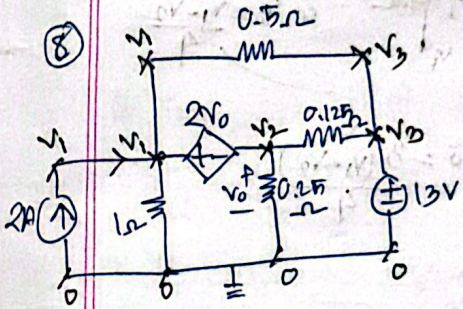
for node V_3,

$\frac{V_3 - V_2}{10} + \frac{V_3 - 0}{10} = 4I_x$

$= \frac{V_2}{5}$

solve all then find. $V_o = V_3$.

⑧



Find V_o and node voltages.

Hint: $V_o = V_2$

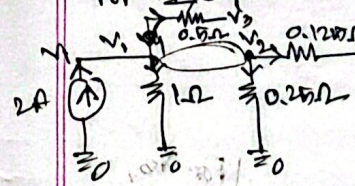
and $V_1 - V_2 = 2V_o$

$\Rightarrow V_1 - V_2 = 2V_2$

$\Rightarrow \boxed{V_1 - 3V_2 = 0}$

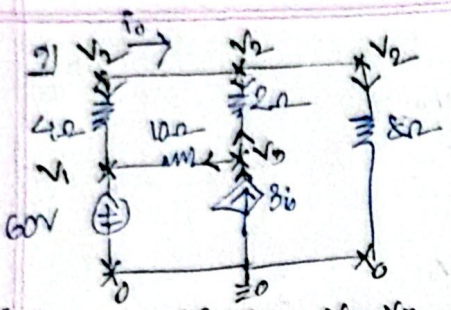
$\boxed{V_3 = 13V}$

For V_1 supernode,



$\boxed{\frac{V_1}{1} + \frac{V_1 - V_2}{0.5} + \frac{V_2}{0.25} + \frac{V_2 - V_3}{0.125} = 2}$

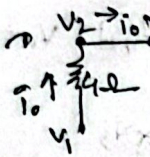
solve boxed eqns to get V_1, V_2, V_3 and then $V_o = V_2$.

2) 

Find is using node analysis.

hint $V_1 = 60V$.

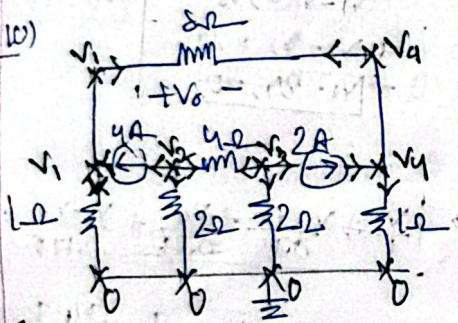
For V_2, $\frac{V_2-V_3}{2} + \frac{V_2-0}{8} + \frac{V_2-V_1}{4} = 0$

Again $i_o = ?$ 

So, $i_o = \frac{V_1-V_2}{4}$

For V_3,

$$\frac{V_3-V_2}{2} + \frac{V_3-V_1}{10} = 3i_o = 3\left(\frac{V_1-V_2}{4}\right)$$

10) 

Find V_o.

hint

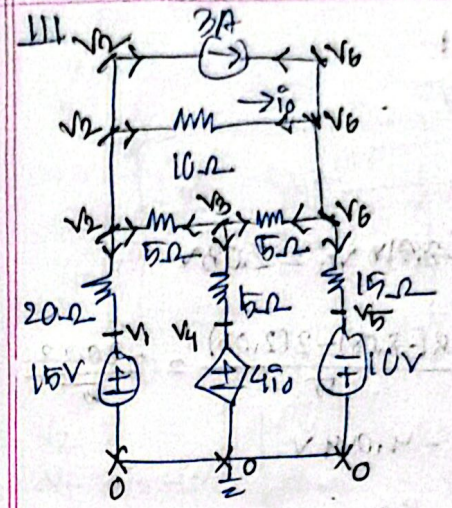
For V_1, $\frac{V_1-V_4}{8} + \frac{V_1-0}{1} = 4$

For V_2, $\frac{V_2-V_3}{4} + \frac{V_2-0}{2} + 4 = 0$

For V_3, $\frac{V_3-V_2}{4} + \frac{V_3-0}{2} + 2 = 0$

For V_4, $\frac{V_4-0}{1} + \frac{V_4-V_1}{8} = 2$

Then calculator.

11) 

Find node voltages.

$V_1 = 15V$, $V_5 = -10V$.

$i_o = \frac{V_2-V_6}{10}$

$4i_o = \frac{2A(V_2-V_6)}{10\,5} = \frac{2V_2-2V_6}{5}$

$V_4 - 0 = 4i_o$

$\Rightarrow V_4 = 4i_o = \frac{2V_2-2V_6}{5}$

For V_2,

$$\frac{V_2-V_1}{20} + \frac{V_2-V_3}{5} + \frac{V_2-V_6}{10} + 3 = 0$$

$$\Rightarrow \frac{V_2-15}{20} + \frac{V_2-V_3}{5} + \frac{V_2-V_6}{10} + 3 = 0$$

$$\Rightarrow V_2 - 15 + 4V_2 - 4V_3 + 2V_2 - 2V_6 + 60 = 0$$

$$\Rightarrow 7V_2 - 4V_3 - 2V_6 = -45$$

For V_3,

$$\frac{V_3-V_4}{5} + \frac{V_3-V_6}{5} + \frac{V_3-V_2}{5} = 0$$

$$\Rightarrow V_3 - V_4 + V_3 - V_6 + V_3 - V_2 = 0$$

$$\Rightarrow -V_2 + 3V_3 - \left(\frac{2V_2-2V_6}{5}\right) - V_6 = 0$$

$$\Rightarrow -5V_2 + 15V_3 - 2V_2 + 2V_6 - 5V_6 = 0$$

$$\Rightarrow -7V_2 + 15V_3 + V_6 = 0$$

For V_6,

$$\frac{V_6-V_5}{15} + \frac{V_6-V_3}{5} + \frac{V_6-V_2}{10} = 3$$

$$\Rightarrow 2V_6 - 2V_5 + 6V_6 - 6V_3 + 3V_6 - 3V_2 = 90$$

$$\Rightarrow -3V_2 - 6V_3 - 2V_5 + 11V_6 = 90$$

$$\Rightarrow -3V_2 - 6V_3 - 2(-10) + 11V_6 = 90$$

$$\Rightarrow -3V_2 - 6V_3 + 11V_6 = 90 - 20 = 70$$

Scanned with CamScanner

So,

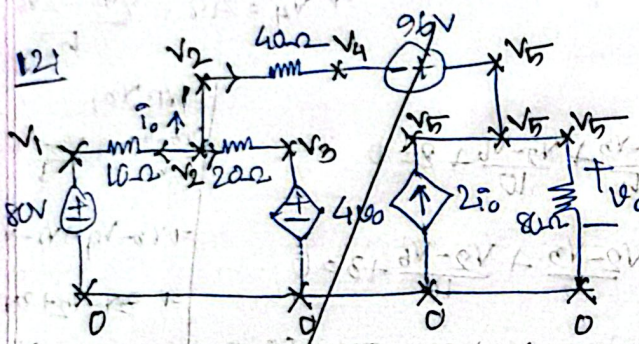
V_2	V_3	V_6	const
7	-4	-2	-45
-7	15	1	0
-3	-6	11	70

$V_2 = -8.08V$, $V_3 = -8.91V$, $V_6 = 2.03V$

$$V_4 = \frac{2V_2 - 2V_6}{5} = \frac{2(-8.08) - 2(2.03)}{5} = \frac{-20.22}{5}$$

$$= -4.044\ V.$$

12)



$i_o = \frac{V_1 - V_2}{10}$ & then V_4 and V_1

$= \frac{V_1 - V_2}{10}$

$2i_o = 2\left(\frac{V_1 - V_2}{10}\right) = \frac{V_1 - V_2}{5}$

for V_1,

$V_1 - 0 = 80 \Rightarrow V_1 = 80$

for V_2,

$$\frac{V_2 - V_1}{10} + \frac{V_2 - V_3}{20} + \frac{V_2 - V_4}{40} = 0$$

for V_3,

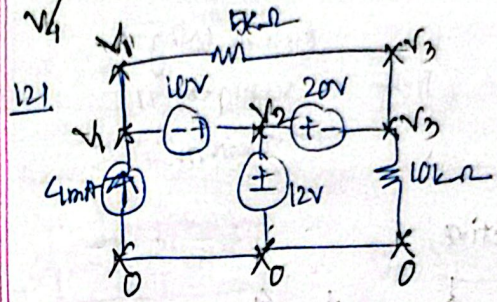
$V_3 - 0 = 4V_o$

Again, $V_o = V_5$

So, $V_3 = 4V_5$

For V_4,

12)



Find node voltages.

$V_2 = 12.$

$V_1 - V_2 = -10$

$\Rightarrow V_1 = -10 + V_2$

$= -10 + 12$

$= +2V.$

$V_2 - V_3 = 20$

$\Rightarrow V_3 = V_2 - 20$

$= 12 - 20$

$= -8V.$